MS&E 226 — Midterm 1 Practice Questions (Generated Set 2): Solutions

Question 1

An online retailer fits OLS to model delivery cost as a function of shipping method, a categorical feature with three levels: Ground (the baseline category), Air, and Express. The model is `cost ~ 1 + shipping`, and R reports: intercept $= 12$, `shippingAir` $= 16$, `shippingExpress` $= 33$ (all in dollars). What does the coefficient on `shippingExpress` represent?

(a) The difference between the mean delivery cost for Express shipments and the mean delivery cost for Ground (baseline) shipments.

(b) The mean delivery cost for Express shipments, in dollars.

(c) The mean delivery cost for Express shipments minus the overall average delivery cost across all three shipping methods.

(d) The change in delivery cost associated with a one-unit increase in "amount of express shipping."

(e) I choose not to answer this question.

Answer: (a)

(a) Correct. With one-hot encoding and Ground as baseline, each non-baseline coefficient is exactly the gap between that category's group mean and the baseline group's mean – here, mean(Express) − mean(Ground) $= 33$.

(b) That would be intercept + coefficient ($12 + 33 = 45$), not the coefficient alone.

(c) The baseline for a one-hot-encoded categorical feature is the **reference category** (Ground here), not the grand mean across all categories.

(d) `shipping` is categorical, not continuous – there's no "one-unit increase" in a dummy variable to speak of; this confuses a continuous-feature coefficient interpretation with a categorical one.

Question 2

You include two highly correlated (but not perfectly collinear) features – years of experience and age – in an OLS model predicting salary. Evaluate the following statements:

(i) The model's predictions (and R^2) are reliable despite the high correlation between the two features.

(ii) The individual coefficients on experience and age can each be interpreted as that feature's "pure," independent effect on salary, unaffected by the correlation between them.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Answer: (a)

(i) is correct. As long as the two features aren't **exactly** collinear, $\boldsymbol{X}^\top \boldsymbol{X}$ remains invertible and OLS still finds a well-defined, SSE-minimizing fitted value – high correlation alone doesn't prevent OLS from fitting (and predicting) well.

(ii) is incorrect. When two included features are highly correlated, OLS can't cleanly separate their individual contributions: the coefficients become unstable (small changes in the data can swing them substantially, even flip their sign), so they no longer have a clean "effect of this feature alone" reading. Collinearity threatens coefficient **interpretability**, not predictive accuracy.

(a) Correct – see the reasoning above.

(b) (i) is also correct, so (ii) alone is not the full picture.

(c) (ii) is not correct, so both cannot be.

(d) (i) is correct, so it isn't true that neither is.

Question 3

You evaluate several candidate models on a validation set and select the model with the lowest validation RMSE as the "winner." You then evaluate that winner on a separate test set that played no role in selecting the winner. Evaluate:

(i) The winner's validation RMSE is an unbiased estimate of its true generalization error.

(ii) The winner's test RMSE is an unbiased estimate of its true generalization error.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Answer: (b)

(ii) is correct, (i) is not. The winner was specifically chosen **because** it had low validation error among the candidates – that selection step means its validation RMSE is optimistic (a form of selection bias), so it systematically underestimates the winner's true generalization error. The test set played no role in choosing the winner, so the winner's test RMSE is an honest, unbiased estimate.

(a) (i) is not correct; see above.

(b) Correct – see the reasoning above.

(c) (i) is not correct, so both cannot be.

(d) (ii) is correct, so it isn't true that neither is.

Question 4

You fit a ridge regression using two features – income (ranging from \$10,000 to \$500,000) and a 0/1 indicator for `has_degree` – without standardizing either feature first. What is the consequence of skipping standardization here?

(a) Because the penalty $\lambda \sum_j \hat{\beta}_j^2$ is applied to the raw coefficient values, and income needs a tiny coefficient (since it's measured in large dollar units) while `has_degree` needs a much larger coefficient for comparable predictive impact, the penalty shrinks the two features' coefficients asymmetrically even if the features are equally predictive.

(b) There's no real issue – ridge regression is scale-invariant by construction, so standardization is purely a convenience, not a requirement.

(c) Skipping standardization makes the fit biased even at $\lambda = 0$, since the unpenalized OLS solution itself depends on feature scale.

(d) Skipping standardization only matters for lasso's sparsity property; ridge's shrinkage behavior is unaffected by feature scale.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct. This is exactly why lecture states you should always standardize before fitting ridge or lasso: the penalty term treats raw coefficient magnitude as "large," but what counts as a large coefficient is purely an artifact of the feature's units.

(b) Contradicts the lecture's explicit standardization requirement – ridge is **not** scale-invariant, precisely because the penalty term is scale-sensitive even though the SSE term alone isn't.

(c) At $\lambda = 0$, ridge reduces exactly to OLS, and OLS's fitted values are scale-invariant (rescaling a feature just rescales its coefficient correspondingly) – the problem is specific to the penalty term, not the base SSE objective.

(d) The standardize-first principle applies to both ridge and lasso equally – both penalties are scale-sensitive in the same way.

Question 5

From a dataset with 500 candidate features, you compute each feature's correlation with the outcome using the **entire** dataset, keep the 10 most correlated features, and then run 10-fold CV using only those 10 features to estimate the generalization error of the final model (which will be fit on all the data using those 10 features). Is the resulting Err_{CV} trustworthy?

(a) No – Err_{CV} will typically **underestimate** the true generalization error, because the 10 features were chosen using information from every fold, including each fold when it is later held out, so the held-out folds aren't fully out-of-sample with respect to the feature-selection step.

(b) No – Err_{CV} will typically **overestimate** the true generalization error, because using only 10 of 500 features makes the model simpler and therefore harder to fit well.

(c) Yes, this is fine, because the outcome variable itself is never used inside any fold's model-**fitting** step – only in the earlier screening step.

(d) Yes, this is fine, as long as $K \geq 5$.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct – this is precisely the "cautionary hypothetical" from lecture: covariate selection is part of the learning algorithm, and it must be redone inside each fold (using only that fold's training data), or the held-out fold's information leaks into the model-selection step applied to it.

(b) Backwards on both the direction and the mechanism – the problem isn't model simplicity, it's that the feature-selection step saw data it shouldn't have.

(c) This is exactly the tempting-but-wrong defense: the outcome **was** used, just one step earlier (to compute correlations for screening), and that's enough to leak held-out information into the folds that are supposed to be excluded from it.

(d) The choice of K has nothing to do with this problem – the issue is leakage of information across the train/held-out boundary, which no choice of K fixes.

Question 6

Compare $K = 5$ vs. $K = n$ (leave-one-out) cross-validation for estimating the generalization error of a model that will ultimately be trained on the **entire** dataset. Evaluate:

(i) Using $K = 5$ instead of $K = n$ tends to **overestimate** the true generalization error of the final (full-data) model, since each fold's model is trained on less data than the final model will be.

(ii) Using $K = n$ always gives a strictly more precise (lower variance) estimate of generalization error than $K = 5$, simply because it uses more folds.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Answer: (a)

(i) is correct: with $K = 5$, each $\hat{f}^{-k}$ trains on only $\frac{4}{5}$ of the data, so it tends to have higher error than a model trained on all of it – inflating Err_{CV} relative to the final model's true error.

(ii) is incorrect: lecture explicitly flags the opposite risk at large K – with $K = n$, the n fold-models differ from each other by only one observation each, so they're nearly identical, which makes Err_{CV} highly **sensitive** to the specific dataset you happened to draw (i.e., high variance if you imagine repeating the whole procedure on a fresh dataset). "More folds" doesn't automatically mean "more precise."

(a) Correct – see the reasoning above.

(b) (ii) is not correct; see above.

(c) (ii) is not correct, so both cannot be.

(d) (i) is correct, so it isn't true that neither is.

Question 7

A screening test is applied to $n = 2000$ patients. Of the 40 patients who actually have the condition, the test flags 30 correctly (and misses 10). Of the 1960 patients who don't have the condition, the test incorrectly flags 40 as positive. What is the **false discovery proportion** (FDP) of this test – the fraction of its positive predictions that are actually wrong?

(a) $\frac{40}{70} \approx 0.57$

(b) $\frac{10}{40} = 0.25$

(c) $\frac{40}{1960} \approx 0.02$

(d) $\frac{40}{2000} = 0.02$

(e) I choose not to answer this question.

Answer: (a)

FP $= 40$ (table says so directly); TP $= 40 - 10 = 30$; total predicted positive $=$ TP $+$ FP $= 30 + 40 = 70$. FDP $= \frac{\text{FP}}{\text{TP+FP}} = \frac{40}{70} \approx 0.57$.

(a) Correct, by the computation above.

(b) This is $\frac{\text{FN}}{\text{FN+TP}} = \frac{10}{40}$, the false negative rate – a rate among actual positives, not among positive predictions.

(c) This is $\frac{\text{FP}}{\text{TN+FP}} = \frac{40}{1960}$, the false positive rate – a rate among actual negatives, not among positive predictions.

(d) This divides by the total sample size instead of by the number of positive predictions, which isn't any of the standard confusion-matrix metrics.

Question 8

Classifier A has AUC $= 0.80$; Classifier B has AUC $= 0.75$, but their ROC curves cross: B's curve lies **above** A's for false positive rates below 0.1, and below A's curve everywhere else. A hospital screening for a dangerous condition is only willing to tolerate a false positive rate of at most 5% (i.e., it will operate strictly within that crossing region). Which classifier should it prefer, and why?

(a) Classifier B – at the hospital's actual operating point (FPR $\leq 5\%$, inside the region where B's curve lies above A's), B achieves a higher true positive rate there, even though A has the higher **overall** AUC. AUC summarizes performance averaged across all thresholds, not specifically at the one operating point that matters here.

(b) Classifier A, because it has the higher AUC, and AUC is always the right criterion for choosing between classifiers regardless of context.

(c) It doesn't matter which one they choose, since both classifiers achieve the same accuracy at any fixed threshold.

(d) Classifier A, because a crossing ROC curve means Classifier B's results are unreliable.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct – a higher-AUC classifier need not give the better tradeoff at the **specific** threshold a decision-maker actually cares about; that's exactly why ROC curves (not just the single AUC number) matter for decision-making.

(b) This is the mistake the question is testing: AUC is a convenient single-number summary, but it compresses away exactly the threshold-specific information a real decision-maker needs.

(c) False – the two classifiers have different TPRs at a fixed FPR (that's what "the curves cross" means); their accuracies at any given threshold generally differ.

(d) Crossing ROC curves are a completely ordinary occurrence between two reasonable classifiers – it doesn't imply either one is unreliable.

Question 9

Which of the following is **not** one of the three steps in the general maximum likelihood estimation (MLE) recipe presented in lecture?

(a) Assume a distributional form for how the data was generated.

(b) Compute (write down) the likelihood of the observed data under that assumed distribution.

(c) Choose the parameter values that maximize that likelihood.

(d) Split the data into training and validation sets to prevent overfitting.

(e) I choose not to answer this question.

Answer: (d)

(a) This is Step 1 of the MLE recipe.

(b) This is Step 2 of the MLE recipe.

(c) This is Step 3 of the MLE recipe.

(d) Correct – this is model **selection/evaluation** methodology (train-validate-test), a separate concern from how MLE fits one given model to data. MLE is a fitting procedure; splitting data is how you'd later choose among different fitted models or estimate generalization error. Conflating the two is a common mistake.

Question 10

Which of the following is correct?

(a) Logistic regression's negative log-likelihood (log loss) is convex in the coefficients, so any local minimum found by an iterative optimizer is guaranteed to be the global minimum; this guarantee does **not** carry over to neural networks, whose loss surfaces are generally non-convex.

(b) Both logistic regression and neural network training are convex optimization problems, since both minimize a loss function using gradient-based methods.

(c) Logistic regression's loss is non-convex, which is exactly why iteratively reweighted least squares is needed instead of plain gradient descent.

(d) Convexity is a property of the dataset, not the loss function, so whether logistic regression's optimization is convex depends on whether the data happens to be linearly separable.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct – this convex vs. non-convex distinction is exactly why logistic regression always finds the (unique) MLE reliably, while neural network training has no such guarantee and can get stuck in different local minima depending on initialization.

(b) Using gradient-based methods doesn't make a problem convex – gradient descent is applied to both convex and non-convex problems; it's a property of the loss surface's shape, not of the optimizer.

(c) Backwards – logistic regression's log loss **is** convex; IRLS is simply one efficient algorithm for solving that particular convex problem, not a workaround for non-convexity.

(d) Convexity here is a property of the **log-loss function** as a function of the coefficients, which holds regardless of whether the data is linearly separable (though a finite MLE may fail to exist in the perfectly-separable case, convexity itself doesn't depend on it).

Question 11

You fit ridge regression on **standardized** features, so that as $\lambda \to \infty$, every feature coefficient is driven to exactly zero and the model's prediction converges to just the intercept (which, for standardized features, equals $\overline{Y}$). Assuming the true population model is **not** simply a constant, what happens to the model's bias and variance as $\lambda \to \infty$?

(a) Both converge to the bias and variance of the constant-mean predictor $\overline{Y}$: variance shrinks to the (generally small but nonzero) variance of the sample mean, while bias converges to a fixed, generally **nonzero** value – the gap between the true conditional mean and the unconditional population mean. Ridge trades away its features' predictive content for stability, landing at a high-bias, low-variance extreme.

(b) Both bias and variance converge to zero, since $\lambda \to \infty$ makes the model maximally regularized and therefore maximally accurate.

(c) Variance converges to zero (the model no longer depends on any feature), but bias stays exactly zero throughout, just as it was for OLS under (A1)–(A3), since ridge never introduces bias.

(d) Both bias and variance increase without bound as $\lambda \to \infty$, since an infinitely large penalty destabilizes the fitting procedure.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct – in the limit, ridge is no longer "ridge regression on features," it's just "predict the sample mean," which is exactly the constant-prediction case from the bias-variance decomposition (zero variance in the population-model sense, generally nonzero bias unless the truth happens to be constant).

(b) Maximal regularization does not mean maximal accuracy – it means maximal **simplicity**, which trades variance for bias; pushed to the extreme, you simply discard all the features' information.

(c) Ridge's zero-bias property under (A1)–(A3) belongs to OLS ($\lambda = 0$) specifically; once $\lambda > 0$ shrinks coefficients away from the unbiased OLS solution, bias is generally reintroduced – exactly the Gauss-Markov tradeoff, taken to its limit.

(d) Backwards – variance **shrinks** (not grows) as $\lambda \to \infty$, since the fitted model becomes **less** sensitive to which particular training sample was drawn, not more.

Question 12

Suppose the population model is $Y = 4X + \varepsilon$, where $X > 0$, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathrm{Var}(\varepsilon \mid X) = \frac{16}{X^2}$ (the error variance **shrinks** as X grows). You have a sample in which X takes many different values, and you estimate the coefficient on X (no intercept) two ways: by OLS, and by WLS with weights $w_i = X_i^2$. Evaluate:

(i) The OLS estimate of the coefficient is unbiased.

(ii) The WLS estimate of the coefficient has lower variance than the OLS estimate.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements (i) and (ii) are correct.

(d) Neither statement (i) nor (ii) is correct.

(e) I choose not to answer this question.

Answer: (c)

Both are correct. Bias depends only on whether errors average to zero at each X ($\mathbb{E}[\varepsilon \mid X] = 0$ here, for both methods), not on how the error variance behaves – heteroskedasticity changes how **noisy** observations are, not where they're centered, so both OLS and WLS are unbiased. Weights $w_i = X_i^2 = \frac{1}{\mathrm{Var}(\varepsilon_i \text{ mid } X_i)}$ are exactly the inverse-variance weights here, which is precisely the choice that gives WLS the minimum possible variance among linear unbiased estimators under heteroskedastic errors (the generalization of Gauss-Markov to the WLS/heteroskedastic setting) – so WLS has lower variance than unweighted OLS.

(a) (ii) is also correct, so (i) alone is not the full picture.

(b) (i) is also correct, so (ii) alone is not the full picture.

(c) Correct – see the reasoning above.

(d) Both (i) and (ii) are correct, so it isn't true that neither is.

Question 13

Suppose the population model is $Y = 10 + 0.5X + X\varepsilon$, where $X > 0$ and ε is normally distributed with mean zero and variance 4, independent of X (so the spread of Y around $10 + 0.5X$ grows with X). Consider the population conditional quantiles of Y given X – the true lines that quantile regression with $\tau = 0.2$ and $\tau = 0.7$ are trying to estimate. Which of the following is correct?

(a) The $\tau = 0.2$ and $\tau = 0.7$ lines are parallel.

(b) The $\tau = 0.2$ line has a larger slope than the $\tau = 0.7$ line.

(c) The $\tau = 0.7$ line has a larger slope than the $\tau = 0.2$ line.

(d) Both lines are the same as the conditional mean line $10 + 0.5X$.

(e) I choose not to answer this question.

Answer: (c)

The τ-quantile of Y given X is $10 + 0.5X + z_\tau \cdot 2X = 10 + (0.5 + 2z_\tau)X$, where z_τ is the τ-quantile of a standard normal. Since $z_{0.7} > 0 > z_{0.2}$, the $\tau = 0.7$ line has the larger slope – the lines fan out as X grows, exactly tracking how the spread of Y around its mean grows with X.

(a) Wrong – parallel lines would mean the spread of Y around the mean is the same at every X, which contradicts the stated model (spread grows with X).

(b) Backwards – $z_{0.2} < 0$, so the $\tau = 0.2$ line's slope $(0.5 + 2z_{0.2})$ is **smaller** than 0.5, not larger than the $\tau = 0.7$ line's.

(c) Correct – see the computation above.

(d) Only the conditional **mean** line is $10 + 0.5X$ – the quantile lines (for $\tau \neq 0.5$) deviate from it by an amount that grows with X, precisely because the noise term is multiplied by X.

Question 14

When using k-nearest-neighbors imputation to fill in missing feature values, which of the following correctly describes the tradeoff in choosing k?

(a) Small k makes the imputed value highly sensitive to the noise in just a few nearby observations (high variance); large k averages over less locally-relevant observations, smoothing out genuine local structure (introducing bias) – directly analogous to the bias-variance tradeoff in using k-NN for prediction.

(b) Small k always introduces more bias, since it ignores most of the dataset; large k always has the lowest variance, since it uses the most available data.

(c) The choice of k for imputation has no connection to the bias-variance tradeoff discussed for prediction, since imputation is a preprocessing step, not a predictive model.

(d) Larger k is always preferable for imputation, since it uses more information and is therefore less likely to be wrong on any single missing value.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct – this is a direct analogue of the usual bias-variance tradeoff in choosing k for k-NN prediction generally: fewer neighbors means more noise-sensitivity, more neighbors means more smoothing/ potential bias.

(b) Reverses which extreme causes which problem – it's **small** k that causes high variance (noise-sensitivity), and **large** k that risks bias (over-smoothing), not the other way around.

(c) Imputation via k-NN averaging is itself a form of local prediction (predicting the missing value from nearby observations), so the same bias-variance logic applies directly.

(d) Ignores the oversmoothing cost of large k – averaging over increasingly dissimilar observations can wash out the genuine local relationship, which is exactly the bias side of the tradeoff.

Question 15

You fit regression trees on the same training set using the same deterministic tree-building procedure, with no complexity penalty, varying only the maximum depth, and use 5-fold cross-validation to estimate generalization error at each depth. Evaluate:

(i) Shallow trees (small maximum depth) tend to have higher bias than deeper trees.

(ii) Cross-validation RMSE is minimized at the deepest tree considered, since deeper trees always fit the training data at least as well.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Answer: (a)

(i) is correct: a shallow tree can only partition the data into a few coarse groups, so it can't capture finer structure in the true relationship – it underfits, i.e. has higher bias.

(ii) is incorrect: it's **training** error that is non-increasing in depth (deeper trees can only fit the training data at least as well), but that's a different claim from CV error. Deeper trees become increasingly sensitive to the specific training sample (higher variance), so CV error typically falls and then **rises** again at large depth – it is generally minimized at some intermediate depth, not the deepest one considered.

(a) Correct – see the reasoning above.

(b) (ii) is not correct; see above.

(c) (ii) is not correct, so both cannot be.

(d) (i) is correct, so it isn't true that neither is.

Question 16

Let $\hat{p}$ be the fraction of "successes" in a sample of n i.i.d. Bernoulli(p) trials (e.g., the fraction of surveyed voters who support a candidate), used as an estimator of the true population proportion p. Using the estimator bias-variance framework (MSE $=$ Bias^2 $+$ Variance, no irreducible-error term), which of the following is correct about $\hat{p}$ as an estimator of p?

(a) $\hat{p}$ is unbiased for p ($\mathbb{E}[\hat{p}] = p$), and its variance is $\frac{p(1-p)}{n}$, which shrinks as n grows – so MSE $= 0 + \frac{p(1-p)}{n} \to 0$ as $n \to \infty$.

(b) $\hat{p}$ is unbiased, but since estimating a proportion involves averaging random (0/1) outcomes, MSE must also include an irreducible error term reflecting the randomness of each individual trial.

(c) $\hat{p}$ is biased toward 0.5, since averaging binary outcomes pulls the estimate toward the middle of its possible range.

(d) The variance of $\hat{p}$ is $p(1-p)$ and does not depend on n, since variance is a population quantity, not a sample quantity.

(e) I choose not to answer this question.

Answer: (a)

(a) Correct. $\hat{p}$ is a sample mean of i.i.d. Bernoulli(p) draws, so $\mathbb{E}[\hat{p}] = p$ (unbiased) and $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$ (population variance of one Bernoulli draw, divided by n, exactly like any sample mean).

(b) This is the Lecture 7 vs. Lecture 8 confusion again: irreducible error belongs to **predicting a new random outcome** Y, not to **estimating a fixed population parameter** p. Here the target is p itself (fixed under the frequentist view), so there's no irreducible-error term, no matter how "random" the individual trials feel.

(c) $\hat{p}$ is an unbiased average of the data – it's centered exactly at the true p, wherever p happens to be, not pulled toward 0.5.

(d) Confuses the variance of a **single** Bernoulli draw ($\mathrm{Var}(Y_i) = p(1-p)$, which indeed doesn't depend on n) with the variance of the **estimator** $\hat{p}$ (the average of n such draws), which does shrink with n like any sample mean's variance.